

Towards an Encyclopedia of Programming Languages Anchored in Software Heritage & CodeCommons

Every Programming Language, with Its Real Programs

Mathieu Acher

CodeCommons plenary meeting · Inria Paris · 28 September 2026



Software Heritage



CodeCommons

PL Catalog: blog.mathieuacher.com/PL-ultimate-llm · github.com/acherm/PL-ultimate-llm

Programming languages + example programs

Browse the repository without dumping everything on one page.

14,242
PL pages

3,842
in-rop (LLM-curated)

3,846
LLM programs

1,242 (9.9%)
PLs with ≥1 ext claim
(of 12,608 in taxonomy)

192
PLs with SWH samples

292
SWH samples

2026-06-12
last build (UTC)

14,242 language pages, cross-referenced with 7 sources + LLM contributions

.m

12 claimants8 primary3 secondaryshared primary (8)

SWH popularity

From the SWH-MSR-ARV dataset (Desmazières, Di Cosmo, Lorentz, MSR 2025; file nb_extensions_alphanum.csv) — one row per (ext, year) in the full SWH archive. Case-aggregated. [citation](#).

46.5M
total occurrences

7.8M
since 2019 (16.7%)

1970–2023
years active

Case variants in archive: .m (46.5M), .M (26.1K) — aggregated above.

Attribution status

This extension is attributed to [GNU Octave](#); [GNU Octave](#); [m4](#); [Mercury](#); [Mercury](#); [Monkey C](#); [Objective-C](#); [Objective-C](#); [Objective-C](#); [octave](#); [octave](#); [Win32 Message File](#); [Wolfram Language](#) as primary by 13 authoritative claims.

Authoritative sources: [Linguist](#) [Pygments](#) [Wikidata](#) [Q169478](#) [Wikipedia](#) · secondary

[Disagree or have a correction? Open a labelling issue.](#)

One page per extension: claimants, SWH popularity, attribution status

Real programs from Software Heritage

6 samples mined from derived_datasets/<date>/contents/*.parquet, byte-verified against the SWH archive. Citation-grade qualified SWHIDs preserved.

MapIt.pm · 571 B · ext .pm · seen 12383x in SWH

via heuristicrule h/linguist/pm/0

swh:1:cnt:238c5c62c8afb3beb1373476b9857b7af32834d;origin=https://github.com/mysociety/fixmystreet;anchor=swh:1:rev:c525f578acef784e08924d1444feff7949b9ace;path=/perl/lib/FixMyStreet/MapIt.pm

Open in SWH · Raw bytes (SWH) · GitHub raw

Show source

98_STOCKQUOTES.pm · 20515 B · ext .pm · seen 11349x in SWH

via heuristicrule h/linguist/pm/0

swh:1:cnt:425464fa7112720ba04fc8d3f525b9de08bde9c;origin=https://github.com/rhop/them-mirror;anchor=swh:1:rev:52ca3800ed14f5e930f7e7d9163ae5bc77aae8b;path=/them/contrib/98_STOCKQUOTES.pm

Open in SWH · Raw bytes (SWH) · GitHub raw

Show source

resultset_overload.t · 543 B · ext .t · seen 1858x in SWH

via heuristicrule h/linguist/t/0

swh:1:cnt:c5ecce87432c0b1e46520cb9b15983f08dc302;origin=https://github.com/st3fan/osx-10.9;anchor=swh:1:rev:9f8135f87c2ba27b1e76a514464b023812eff645;path=/CPAN/Internat-148/DBIx-Class/t/resultset_overload.t

Open in SWH · Raw bytes (SWH) · GitHub raw

Show source

IssuingRules.pm · 7958 B · ext .pm · seen 1695x in SWH

via heuristicrule h/linguist/pm/0

swh:1:cnt:6023671522ebe2915fc8348a5353e5d2dab14084;origin=https://github.com/laurenthd1/koha-bibliore;anchor=swh:1:rev:489b948a84a9efeba23e983ca86942cc2ec439d;path=/C4/IssuingRules.pm

Real programs from SWH: qualified SWHID + the rule that attributed them

Extension review queue

Ranked queue of file extensions present in Software Heritage that need a label. Sorted by priority — highest = most files in SWH and least already claimed by a PL in our taxonomy.

4,556
in queue
(popular ≥ 10.0K occurrences & not well-attributed)

2,805
orphans
(zero evidence anywhere)

1,223
popular exts skipped
(≥ 10.0K occurrences & already well-attributed)

1,803
well-attributed exts
(whole taxonomy, any popularity)

1,994
total exts
in taxonomy

11
manual labels
submitted so far

Extension	SWH	Last	#Claim	Current claimants	Auto-suggested	Samples	Orphan?	Action
.map, .MAP	87.8M	2023	0	—	Build artifact			help this
.log, .Log, .LOG	84.2M	2023	0	—	Build artifact			help this
.lock	61.4M	2023	0	—	lock/cache			help this
.cache, .Cache	46.6M	2023	0	—	lock/cache			help this
.info, .INFO	46.5M	2023	0	—	Build artifact			help this
.flat	39.3M	2023	0	—	unknown			help this

A review queue: 4,556 popular extensions waiting for a label

What the encyclopedia offers

Features of the current prototype

Browse & search

- ▶ 14,242 language pages, A–Z, search by name or alias
- ▶ filters: has an SWH program, has an LLM program, number of sources
- ▶ one page per source: what is only in Esolang?

A page per language

- ▶ which sources attest it
- ▶ Wikipedia / Wikidata facts: designer, year, influences
- ▶ claimed extensions, with source, strength and SWH counts
- ▶ real programs from SWH: qualified SWHID + the rule that fired

Tools & agents, three jobs

- ▶ **find languages**: 7 catalogs, Wikidata candidates, LLM agents
- ▶ **label programs**: PLI tools (Linguist rules, Pygments, Synid), LLM judges, human reviews
- ▶ **find programs**: SWH mining per extension, sample requests

A page per extension

- ▶ claimants and attribution status (well / weakly attributed, polysemous, orphan)
- ▶ SWH popularity, case variants (`.R` vs `.r`)
- ▶ Linguist disambiguation rules

Contribute & review

- ▶ add a language, add a program, label an extension, request SWH samples
- ▶ forms → GitHub issues → merged, auditable
- ▶ review queue, candidates from Wikidata (391)

Transparency

- ▶ every fact is a file in git, every claim has a source
- ▶ statistics: sources, models, agents
- ▶ recent submissions, one-line revert

The dream: every language, with its real programs

The big picture

A **comprehensive, living encyclopedia** of programming languages, **together with their programs in Software Heritage**, from ENIAC to last week's DSL.

- ▶ **Exhaustive**: history included (HOPL, Esolang, PLDB, Linguist...)
- ▶ **Anchored in code**: citable SWHIDs, or a reasoned evidence of absence
- ▶ **Traceable**: every claim says who asserts it, and why
- ▶ **Living**: humans, AIs and automatic procedures add, correct, validate
- ▶ **Sustainable**: carried by SWH × CodeCommons, not a side project

Every language with its programs, every program with its language.

COBOL 1959 · CODASYL

🕒 **Lineage** FLOW-MATIC, COMTRAN, FACT... → COBOL → PL/I (HOPL)

✓ **Sources** PLDB · Linguist · Pygments · Wikidata

✓ **Extensions** .cob .cbl .CBL .cpy, case matters

✓ **Programs** ≈150,600 genuine files, 6,278 repos, SWHIDs

🕒 **Usage over time** raw counts, COBOL extensions



○ **Dialects, domains, papers** at archive scale: to build

✓ available today · 🕒 partial · ○ to build

Why now, and why it matters

✓ The right time

- ▶ **Software Heritage exists:** the largest archive of source code, with citable identifiers, and CodeCommons to analyse it at scale. A unique opportunity
- ▶ **The technology is here:** PLI tools (Linguist, Synid, Tree-sitter), LLMs to propose and annotate, review and collaboration tools
- ▶ → we can map every language to its programs, and every program to its language

✗ The status quo

- ▶ **Many ad-hoc attempts:** HOPL, PLDB, Esolang, Wikipedia lists, Linguist, Rosetta Code... each with its own idea of "a language"
- ▶ **Loosely connected:** no shared identifiers, names that don't match, no link to real code
- ▶ **Fragile:** one-person efforts freeze or go offline (HOPL: frozen around 2005, often down)

Heritage

what ALGOL, COBOL or Icon code really looked like

Science of programming

adoption, evolution and death, per language

AI & LLMs

benchmarks and data for low-resource languages

Tools & SWH

search and statistics by language; ground truth for PLI

Legacy & industry

where is the COBOL, RPG, PL/I code, in which dialects?

Education & citation

real, citable programs (SWHIDs) in papers and textbooks

Where are all the Icon programs?

Reality check

Icon (Griswold, 1977): generators, goal-directed evaluation, string scanning.

- ▶ **HOPL** knows its family: 21 relations (SNOBOL4, SL5, Unicon, Python...)
- ▶ **5 catalogs** list it (PLDB, Pygments, Wikipedia, Rosetta Code, Wikidata)
- ▶ **An LLM** (Llama 3.1 70B) added it, with a program that defines `main` three times
- ▶ **0** verified Icon program in the catalog so far

	<code>.icn</code>	<code>.icon</code>
files in SWH	32,257	19,184
Pygments says	Unicon	Icon
what they really hold*	Icon and Unicon sources (the official Icon repository alone: ~1.2K)	mostly not Icon: icon-theme metadata, icon images; the Icon code comes mainly from one Rosetta Code dump

* GitHub code search, Sept. 2026: ~9.3K `.icn` files; ~56K `.icon` files, of which ~16.5K icon-theme metadata and ~1.3K with Icon code (667 in one Rosetta Code dump).

Finding Icon's programs needs **automated tools and supervised methods** (which extension? which content? Icon or Unicon?) that **scale to all of Software Heritage**: a challenge, and an opportunity.

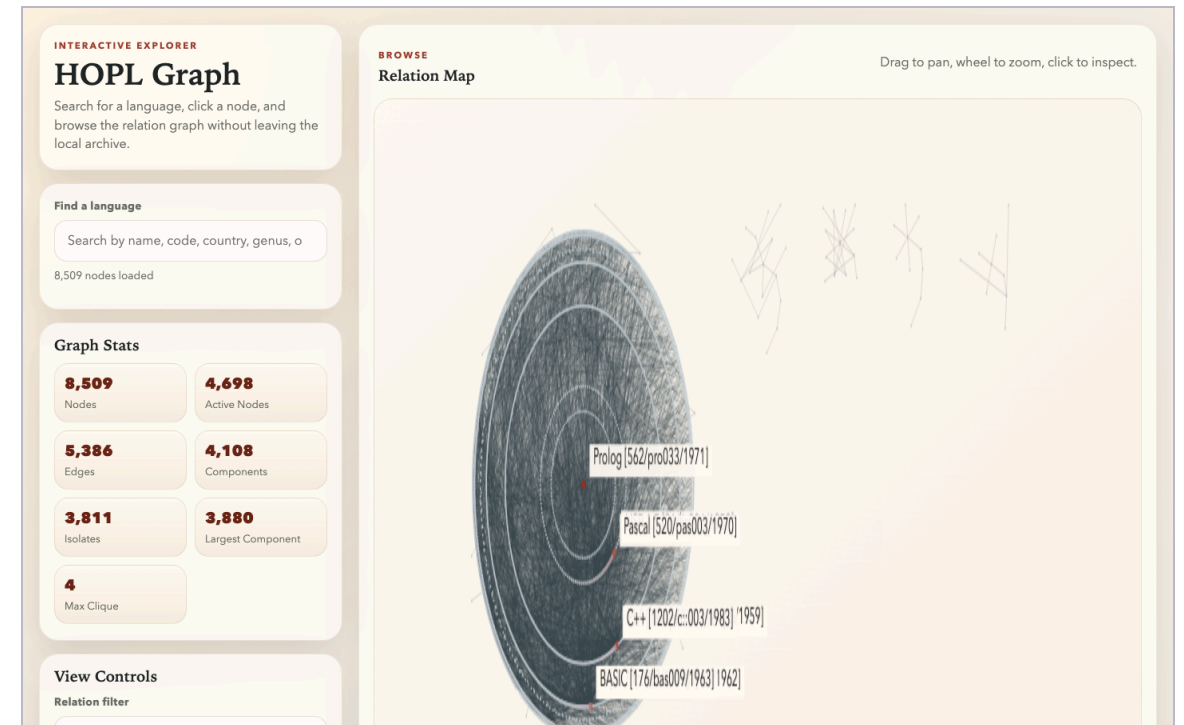
1. Four mirrors, no code

HOPL · PL-ultimate · PL-ultimate-llm · SWH file extensions

HOPL: a genealogy, frozen in 2005

Mirror 1 · history

- ▶ hopl.info: *the* History of Programming Languages, but often **down**, and "please do not copy"
- ▶ Scraped **politely** with Codex: one request at a time, long random delays, contact e-mail in the user agent, resumable, raw HTML archived
- ▶ → **8,509 languages**, **5,386 edges**, 165 relation labels (*Evolution of, Influence, Dialect of, Compiled to...*)
- ▶ Most connected: **Prolog**, FORTRAN IV, PL/I, LISP 1.5
- ▶ 1 language dated after 2004: no Rust, Go, Swift, Kotlin
- ▶ No code



Source	#	Its view of "a language"
Esolang	6,797	esoteric, experimental, jokes
PLDB	5,288	curated metadata
Linguist	878	detectable on GitHub
Wikipedia	769	encyclopedic article
Hyperpolyglot	670	Linguist, in Rust
Pygments	645	someone wrote a lexer
Rosetta Code	640	solved toy tasks

12,608

entities in the union (11,963 in the first build;
+ source updates, Wikidata)

- ▶ **88%** appear in **one** source only
(Esolang 6,366, PLDB 4,034)
- ▶ Near-duplicates survive: `Python` / `py` / `Python` (programming language)
- ▶ Extensions for only ~**10%** of them

PL-ultimate-llm: LLMs contribute through commits

Mirror 3 · what LLMs know

One turn = one commit: read the list, propose a real language **not yet in the list**, give an evidence URL and a program; git hooks validate; trailers record which model did it.

3,862 languages

3,846 commits

since Oct. 2025

Claude 3,392

Grok 246

Gemini 144

Llama 48

GPT 32

DeepSeek 9

	in the LLM list	not in it
in ≥ 1 of the 7 sources	2,439	10,400
in none of them	1,403	—

= the 14,242 pages. Icon sits in the top-left cell: PLDB, Pygments and Rosetta Code had it, the LLM list did not.

Commit 1a1c557

FAMILIAR-project committed on Oct 22, 2025

turn: add Icon (+1 program)

List-Digest: 95f48d04

Model: meta-llama/llama-3.1-70b-instruct

Temperature: 0.4

main

7 files changed

data/pl_list.txt

@@ -53,6 +53,7 @@ HCL

53 53 Hoon

54 54 Humbug

55 55 Hy

56 + Icon

56 57 Idris

57 58 Io

58 59 J

languages/Icon/meta.json

@@ -0,0 +1,8 @@

1 + {

2 + "name": "Icon",

3 + "aliases": [

4 + "Icon Programming Language"

5 +],

6 + "evidence_url": "https://en.wikipedia.org/wiki/Icon_(programming_language)",

7 + "added_at": "2025-10-22T15:15:16Z"

github.com/acherm/PL-ultimate-llm/commit/1a1c557

Do LLMs invent languages? We checked the evidence

All 3,842 evidence links of the LLM list, tested in September 2026

- ▶ **2,439** (63%) are also in one of the 7 sources; of the **1,403** others, **277** are in HOPL or Synid
- ▶ **Evidence links are fragile**: **28%** of the Wikipedia links point to a page that does not exist (**40%** for LLM-only names); others land on the wrong page (Brython → *Ancient Celtic people*)
- ▶ A missing page $\neq$ a fake language: *Aardappel* (*programming language*) has no page, yet Aardappel exists
- ▶ **465** names stay unverified: broken or off-topic evidence, no corroboration

Web check of 40 random unverified names

36 are real languages: 27 as named, 9 under another name (Gringo, SPU assembly, ObScript...). Only their evidence link was wrong.

2 are real, but not languages: NATJ (a Java binding library), Brython (a Python 3 implementation)

2 not found, likely invented: BASIC-D, Oxymoron

$\approx 5\%$ of the 465 (95% CI 1–17%): about **23 invented names in 3,842** (0.6%, at most 2%).
LLMs invent evidence far more often than languages.

Roberto's CSV (SWH-MSR-ARV: Desmazières, Di Cosmo, Lorentz, MSR 2025): **2.96 M** extensions, occurrences per year

Extension	Occurrences in SWH	Claimed by
<code>.m</code>	46.5 M	12 claimants: Octave/MATLAB, Objective-C, Mercury, m4, Wolfram...
<code>.R</code> · <code>.r</code>	21.5 M · 1.7 M	R (case-folding once lost 21.5 M files)
<code>.CBL</code> · <code>.cbl</code>	192 K · 57 K	COBOL
<code>.icn</code> · <code>.icon</code>	32 K · 19 K	Unicon · Icon (Pygments)
<code>.fsf</code>	18 K	claimed by no language (in fact an embedded DSL)

Extensions, not languages. A count is not a program.

Four mirrors, one blind spot

	names languages	lineage	extensions	real programs	maintained
HOPL	✓ 8,509	✓ 5,386 links	✗	✗	✗ frozen ~2005
PL-ultimate	✓ 12,608	🟡 infoboxes	🟡 10% of names	✗ toy tasks	✓ upstream
PL-ultimate-llm	✓ 3,862	✗	✗ rarely	🟡 generated	✓ agents, forms
SWH extensions	✗ suffixes only	✗	✓ 2.96 M	🟡 real, unattributed	✓

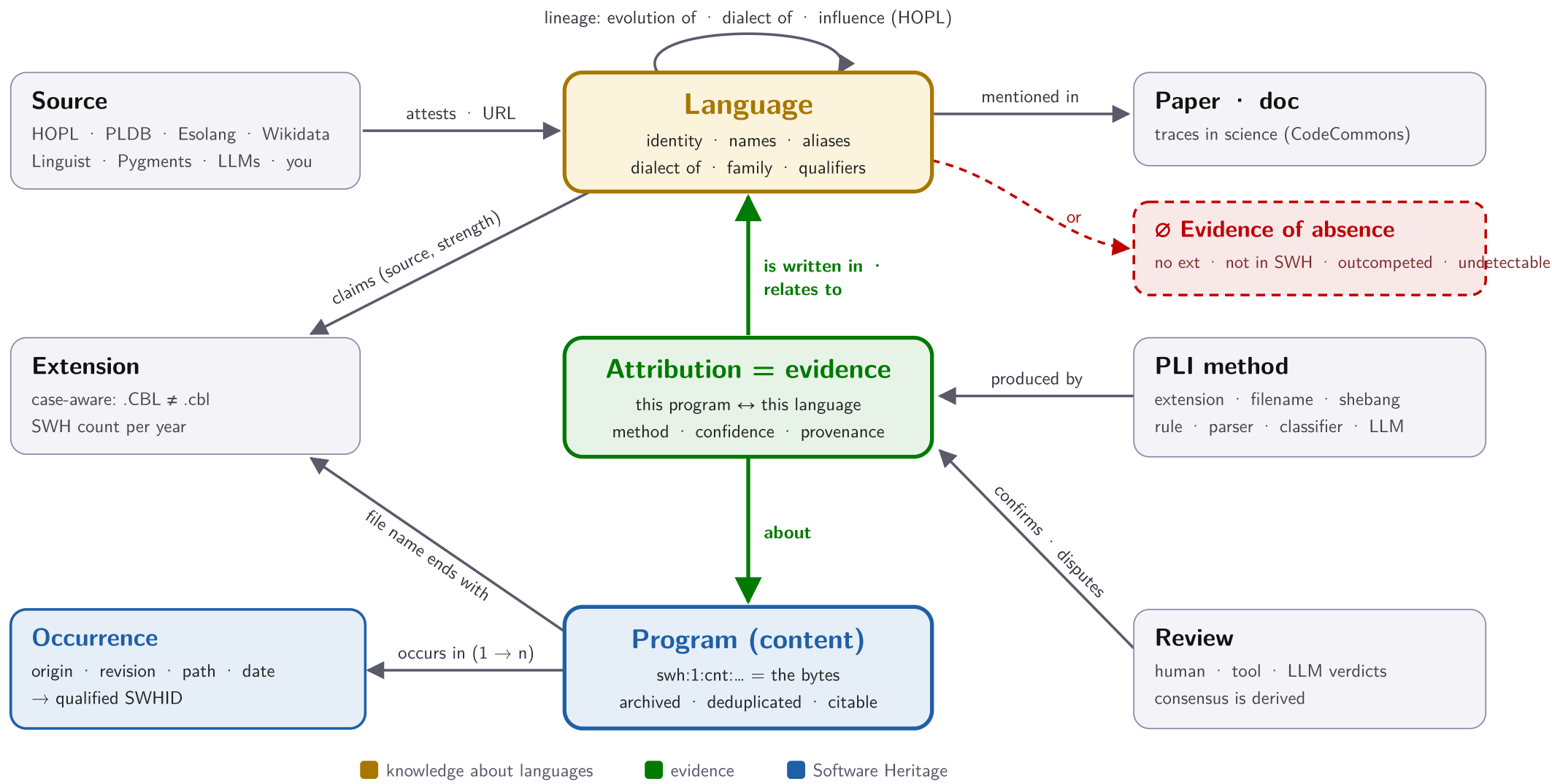
No row has both **names languages** and **real programs**: nothing links a language to the programs that exist.
"Give me 10 real, non-toy **Mercury** programs, with citable provenance": today, impossible.

2. The vision, made concrete

An ontology, a prototype, and the identification problem

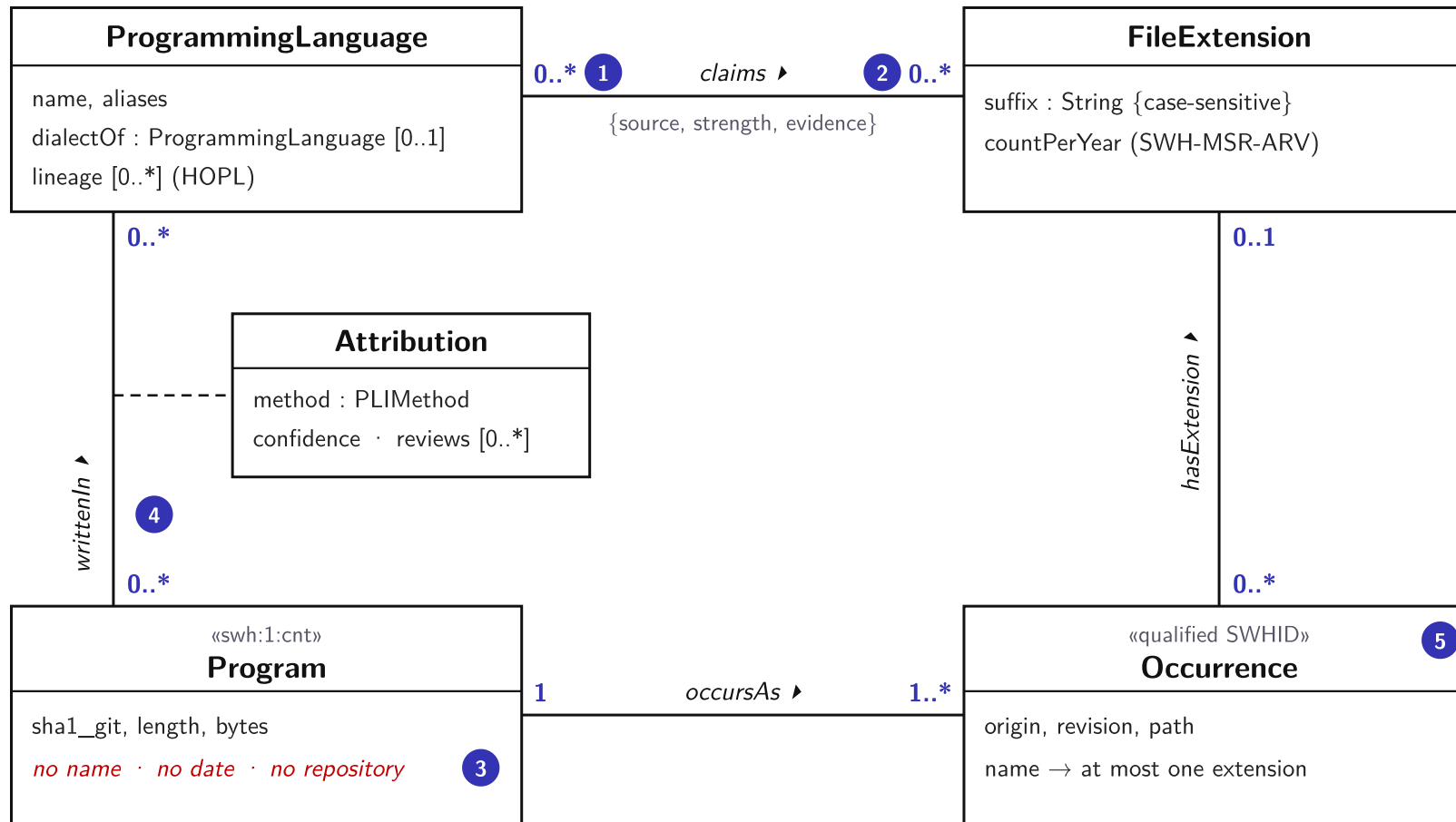
Languages, evidence, programs

The ontology



PL, extension, program: it's all in the cardinalities

A UML view of the core concepts, as seen from Software Heritage



1 A language claims 0..* extensions:
~90% of catalogued languages claim none (esoteric, historical, no files); a few claim 10+ (Python, Ruby).

2 An extension is claimed by 0..* languages: 0 for most (88% of extensions with $\geq 1,000$ files in SWH: `.map`, `.log...`); * is polysemy (`.m`: 12 claimants).

3 Names live in directories: an SWH program is just bytes, so the extension belongs to each occurrence, and one program occurs in many places.

4 writtenIn is established per program, with a method and a confidence, never inferred from a claim: 46% of `.cbl`/`.CBL` files are not COBOL.

5 Counting = choosing a class: programs (by file) or origins (by repo); a new revision is a new program.

Two gaps, in numbers

Languages without extensions, extensions without languages

Languages without extensions

90% of the 12,608 catalogued languages (11,366) claim **no** file extension: invisible to extension-based detection

Extensions without languages

88% of the 18,120 extensions with $\geq 1,000$ files in SWH (15,957) are claimed by **no** language; 56% of those with $\geq 100,000$ files

Help label them: `/review/extensions/` lists **4,556** popular extensions ($\geq 10,000$ files) without a claimant, 2,805 of them with no evidence at all. **11** labels so far.

Extension review queue

Ranked queue of file extensions present in Software Heritage that need a label. Sorted by priority — highest = most files in SWH and least already claimed by a PL in our taxonomy.

4,556
in queue
(popular $\geq 10.0K$ occurrences & not well-attributed)

2,805
orphan
(zero evidence anywhere)

1,223
popular exts skipped
($\geq 10.0K$ occurrences & already well-attributed)

1,803
well-attributed exts
(whole taxonomy, any popularity)

1,994
total exts
in taxonomy

11
manual labels
submitted so far

Extension	SWH	Last	#Claim	Current claimants	Auto-suggested	Samples	Orphan?	Action
<code>.map, .MAP</code>	87.8M	2023	0	—	build-artifact			label this
<code>.log, .Log, .LOG</code>	84.2M	2023	0	—	build-artifact			label this
<code>.lock</code>	61.4M	2023	0	—	lock/cache			label this
<code>.cache, .Cache</code>	46.6M	2023	0	—	lock/cache			label this
<code>.info, .INFO</code>	46.5M	2023	0	—	build-artifact			label this
<code>.flat</code>	39.3M	2023	0	—	unknown			label this

PLI, formally: learn f , one iteration at a time

Conceptualization

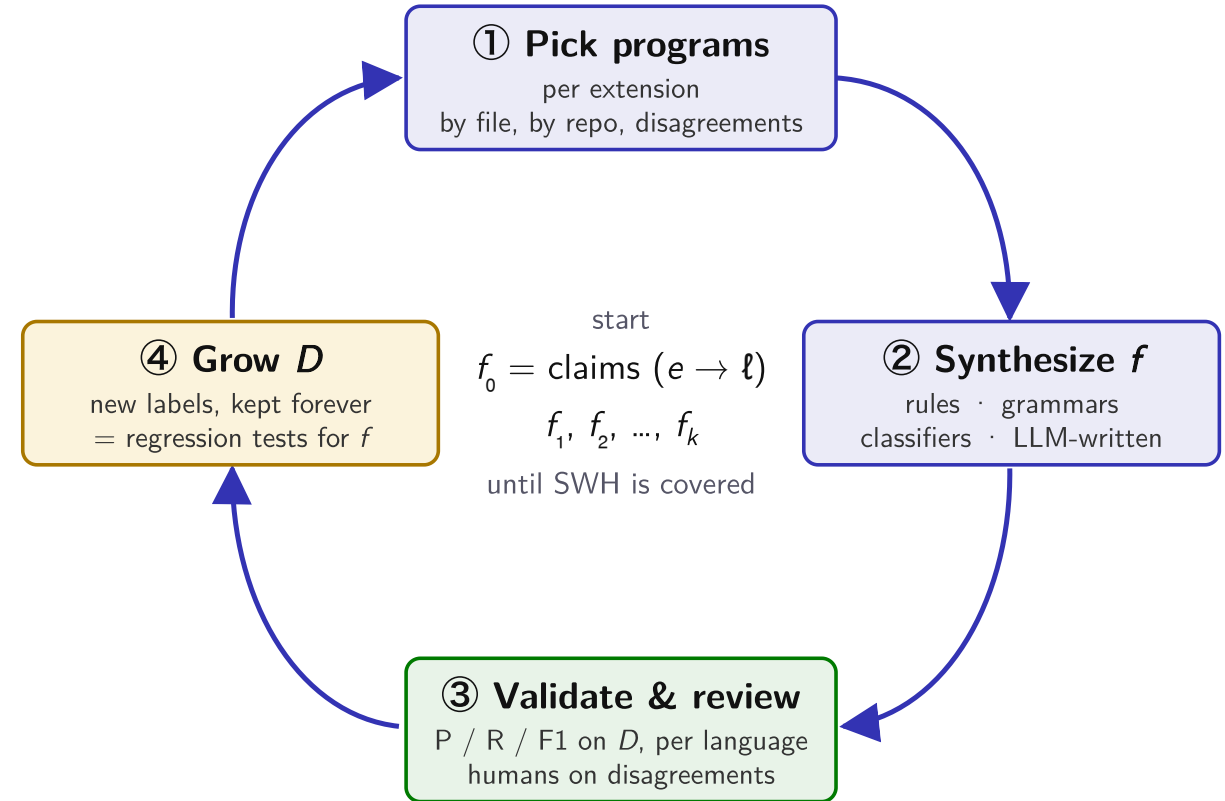
Definition (programming language identification)

- ▶ **Languages**, the labels: $L = \{\ell_1, \dots, \ell_n\}$, $n \approx 12,600$, plus $\perp$ (not a language, unknown)
- ▶ **Extensions**: $E = \{e_1, \dots, e_m\}$, $m \approx 2.96$ M in SWH
- ▶ **Programs**: P , the SWH contents; p occurs with extensions $\text{ext}(p) \subseteq E$
- ▶ **Claims**: $C \subseteq L \times E$, prior knowledge (catalogs, PLI tools)

Goal: find $f: P \times E \rightarrow 2^L$ such that $f(p, e) = \text{writtenIn}(p)$: usually one ℓ , $\emptyset$ for $\perp$, several for embedded languages

Knowledge: labelled examples $D = \{(p, e, \ell)\}$, claims, rules, grammars, models

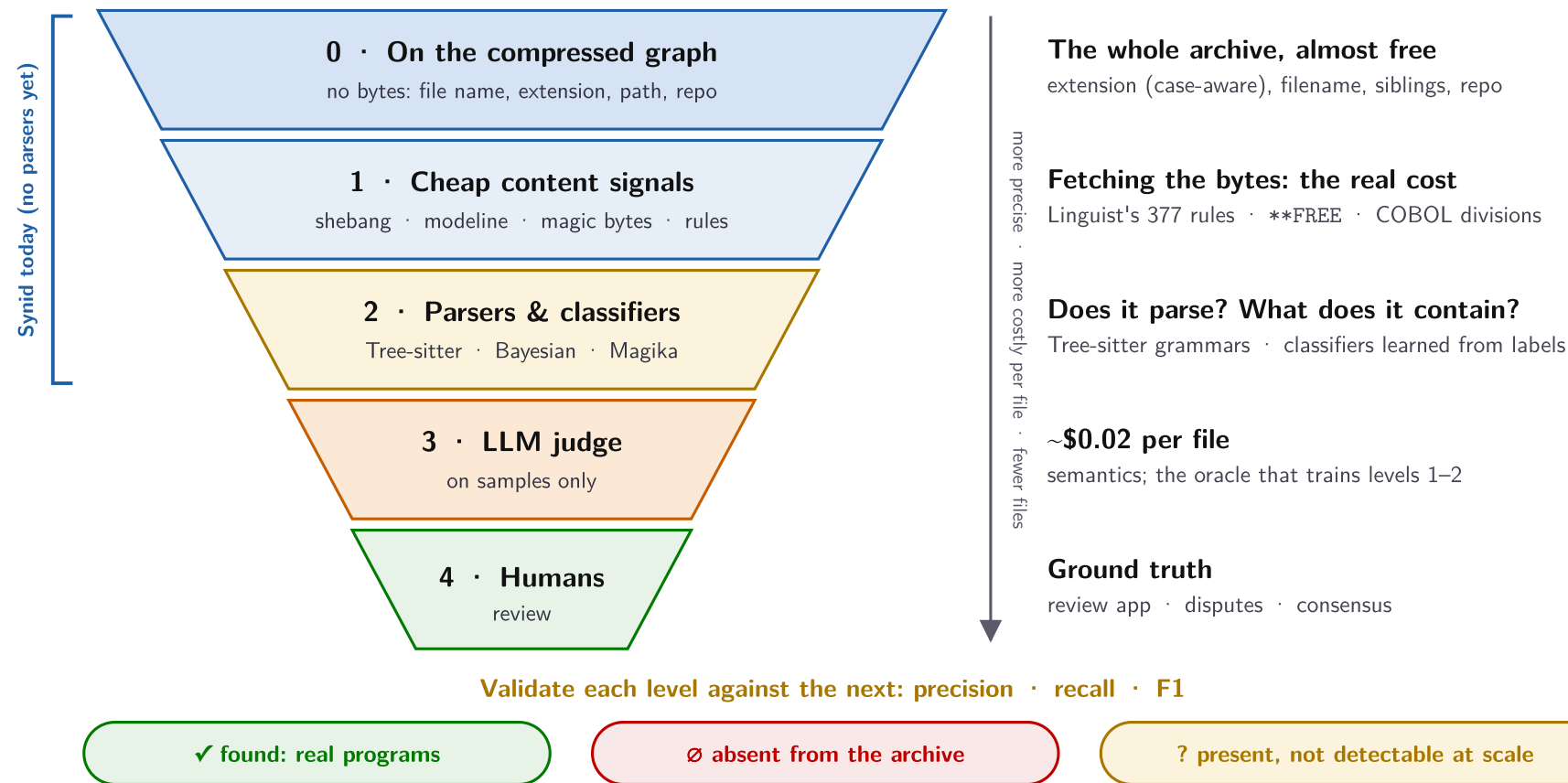
Quality: precision, recall, F1 of f on D , per language and per extension



Not a chicken-and-egg problem (programs to learn f , f to find programs), but an **iterative process**: bootstrap from claims and a few labelled programs; every validated label becomes a **regression test**; examples and knowledge accumulate **across all of SWH**.

Which language is this file in? A multi-level funnel

Programming language identification (PLI)



PLI tools know ~**1,100** syntaxes, Tree-sitter several hundred grammars; we catalogue ~**12,600** languages, ~**90%** with no known extension. Difficult, and fascinating.

Ongoing work, DiverSE team (Rennes): Caroline Landry, Mathieu Acher, Axel Amour N'cho, Baptiste Mehat, Bignon Lokonon, Corentin Ollivier, Guillaume Claudic, Stephan Kunne

3. Zoom in

What is *actually* in file extension X, in all of Software Heritage?

One extension, all of it

A complementary, MSR-style contribution

Why? Low-resource languages:

- ▶ understand **real usage** and the ecosystem
- ▶ build **benchmarks, fine-tune** LLMs
- ▶ find **interesting repositories**

Input (thanks Roberto, Valentin, Baptiste): *every* SWH file with extension X + **its origin**, extracted from the graph with CodeCommons tools

SWH » GitHub, but reading it takes method.

1. **Sample**: by file *and* by repository
2. **Fetch** bytes by SWHID (cached)
3. **Indicators**: cheap, deterministic
4. **LLM-as-judge**: enum-constrained
5. **Reclassifier**: zero-cost, validated vs. the judge
6. **Origins**: provenance, concentration
7. **Human review**: all labels side by side

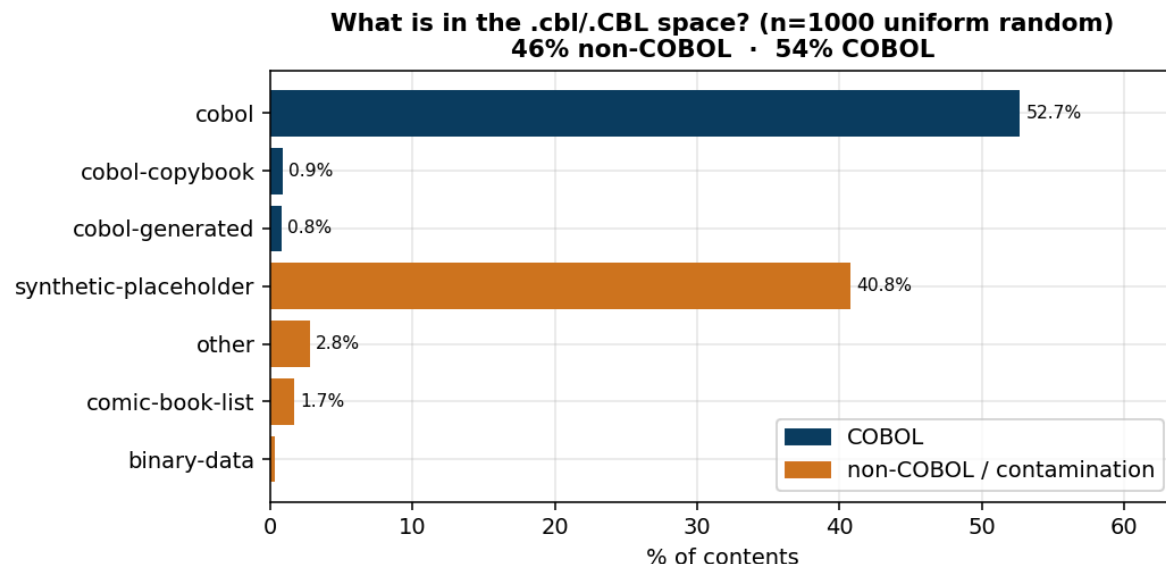
~70% of the toolkit is reused from one extension to the next

Three extensions, three stories

	<code>.cbl</code> · <code>.CBL</code>	<code>.fsf</code>	<code>.rpgle</code>
unique files · repos	276,831 · 6,278	21,802 · 757	13,002 · 534
really the language?	54% COBOL	an embedded DSL (FSL FEAT, in Tcl)	99.5% ILE RPG
what else is in there	one GitLab test fixture (40%), comic-book lists	GLSL shaders, DataLad pointers	parsers <i>for</i> RPG: 21% of all files
the story	contamination, case matters	fMRI analysis designs; polysemy	the sampling frame decides

Next: `.m` (MATLAB/Octave/...), and any extension you care about.

Lesson 1: an extension is a claim, not a fact



- ▶ **46%** of `.cbl`/`.CBL` files are **not COBOL** (95% CI 42.5–48.7)
- ▶ ~110,000 files say *"This is cobol file number N"*: one GitLab repo, a test fixture for huge trees
- ▶ `.CBL` $\neq$ `.cbl`: mainframe enterprise vs learners (+ Calibre comic-book lists)
- ▶ ...yet **93%** of the repositories do contain COBOL

/WBC_97464_FOO.CBL

swlh:1:cnt:4c699ec09243ebef9b1577805c11e422f24977 · 32 bytes · [1k](#) · [SWH](#)

```
This is cobol file number 97464
```

ORIGIN

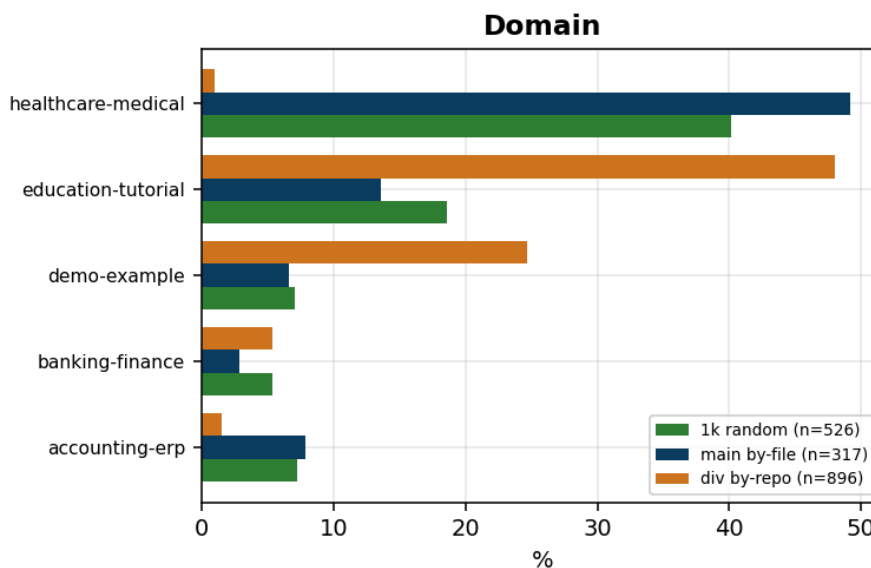
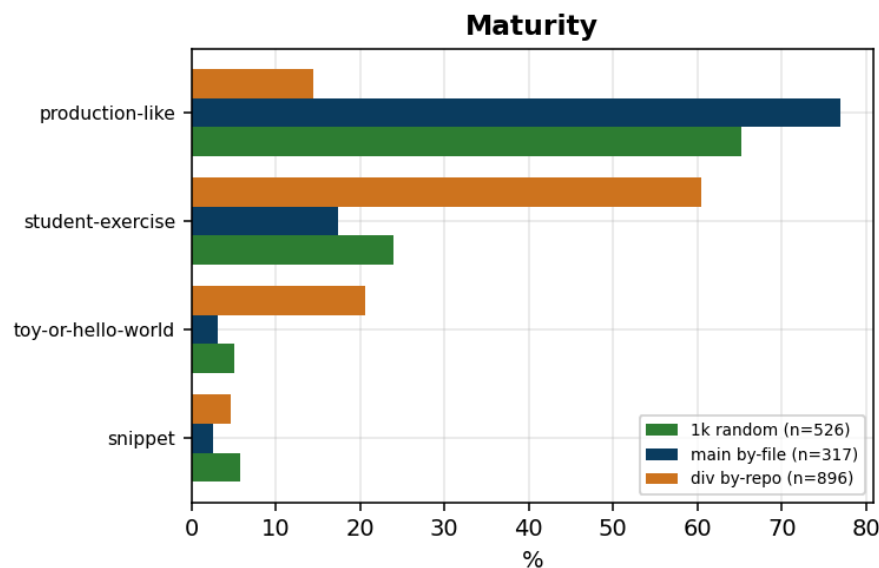
graph
<https://gitlab.com/tbetestpublic/repo-with-many-files-in-one-tree.git> gitlab.com
source/WBC_97464_FOO.CBL · changing-commits · 2022-01-15T13:51:44Z [SWH](#)

LABELS (2 IS-COBOL VOTES · UNANIMOUS)

reclassifier	non	synthetic-placeholder
division-gate	non	n_divisions=0

Lesson 2: how you count changes the story

Sampling frame: random (file) vs curated by-file vs by-repo

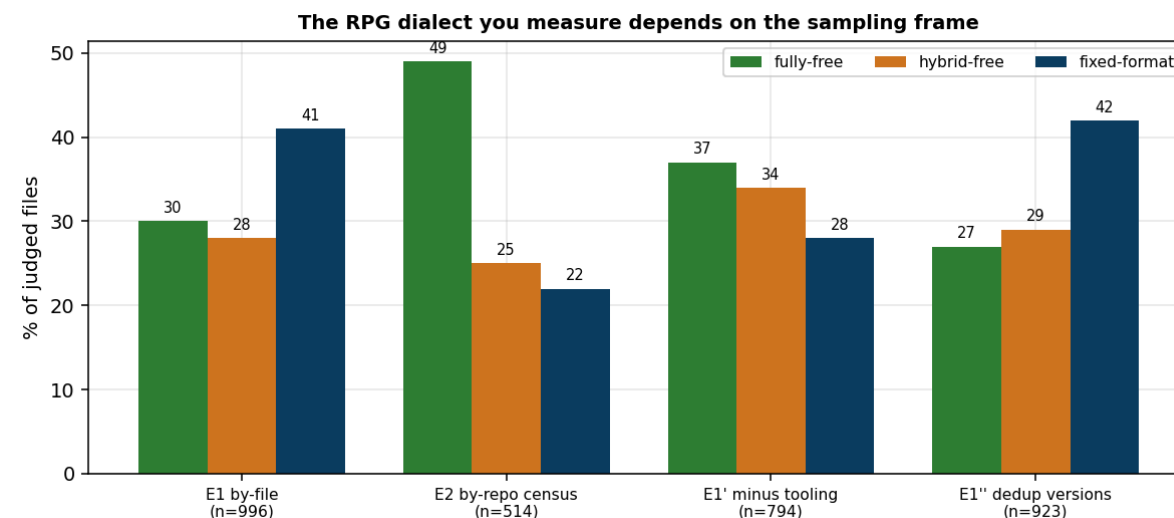


- ▶ **by file:** 77% production-like, 49% healthcare (one system, JMA ORCA, hundreds of files)
- ▶ **by repository:** 14% production-like, **60% student exercises**, median 48 LOC (vs 596)

- ▶ **71 repositories** (1.1%) hold 80% of all COBOL-extension files
- ▶ "Typical **file**" ≠ "typical **project**": report both (same story for RPG, next)

.rpgle: a clean extension, a biased archive

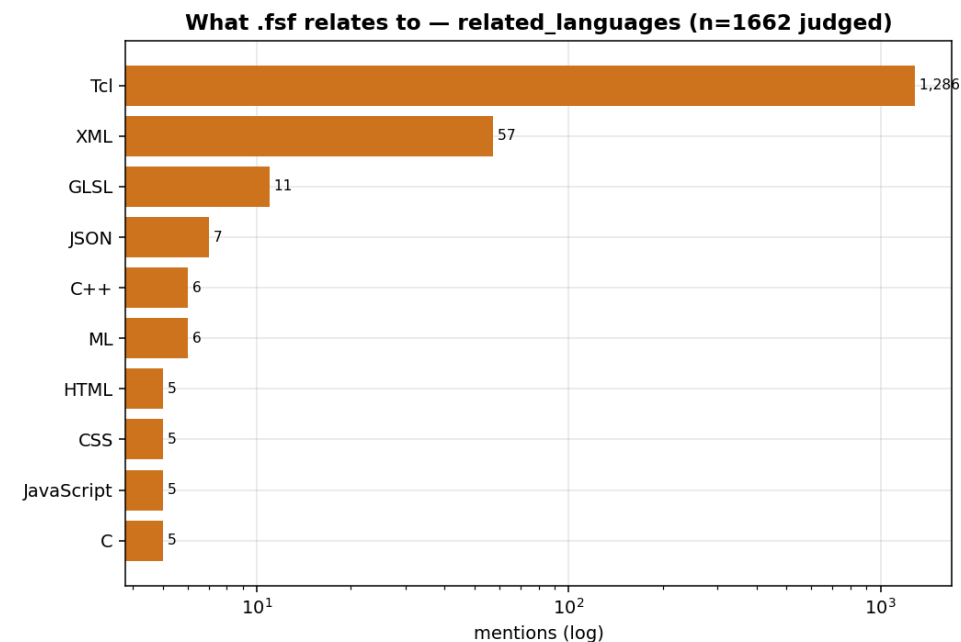
- ▶ **IBM RPG** (1959), still in production on IBM i;
.rpgle = ILE RPG / RPG IV
- ▶ **13,002 files** · **534 repos** · **99.5%** genuine
RPG: an honest extension
- ▶ One language, **three formats**: fixed (column-based), hybrid, fully-free (**FREE on line 1)
- ▶ **Which RPG?** Fixed-format: **41%** by file, **22%** by repo. Fully-free: 30% → **49%**
- ▶ Why: three **parsers for RPG** (an interpreter, two ANTLR grammars) hold **21%** of all .rpgle, 92% fixed-format test fixtures
- ▶ File versions (35% of contents) do **not** bias: 41% → 42%



The archive's picture of RPG is shaped by the tools built to parse RPG.

Lesson 3: an LLM judge is a useful annotator, not an oracle

- ▶ **Enum-constrained** structured outputs (free text fragmented into 30+ spellings)
- ▶ Ask "what **is** this content, what does it **relate to**?", not "is it COBOL?"
- ▶ ~\$0.02 per file: a study costs tens of dollars. **All of SWH: no way**
- ▶ → bootstrap **cheap classifiers**: COBOL reclassifier P 1.00 · R 0.79 · F1 0.88 vs. the judge
- ▶ RPG: the judge contradicts the literal ****FREE** directive on **9.1%** of files, one-sided (132 vs 2)
- ▶ **Code decides lexical facts, the LLM semantic ones, humans arbitrate**



From case studies to a method, and back

An MSR contribution

A general methodology for "what is in extension X?"

- ▶ **sampling**: by file, by repo, stratified... which question does each answer?
 - ▶ **annotating**: LLM vs. PLI tools vs. ad-hoc/ML classifiers; the label problem
 - ▶ **validating**, iterating until *all* files of an extension are covered
- + **domain** contributions: COBOL (batch, CICS...), neuroimaging, IBM i, `.m`

What it gives the encyclopedia

- ▶ a **case-aware** extension → language mapping (`.CBL` ≠ `.cbl`, `.R` ≠ `.r`)
- ▶ **polysemy** made explicit: `related_languages`, `expressed_in`
- ▶ **validated cheap classifiers** → evidence at archive scale
- ▶ evidence of absence, **measured**

Encyclopedia = **breadth** (12K languages, a few programs each). Zoom-in = **depth** (one extension, its whole population).

4. Making it real

An official, sustainable SWH × CodeCommons initiative

Contributions: trace everything, keep it discussable

Today's workflow is a prototype; the principle is what matters

Contribution	How	Lands as
Propose a language	web form → GitHub issue <code>pl-add</code>	PR, merged on arrival
Label an extension	form on <code>/ext/<x>/</code> → issue <code>ext-review</code>	accepted by default → <code>ext_claim</code>
Ask for real programs	"Request SWH samples" → issue <code>sample-request</code>	targeted SWH mining
Submit one file	<code>submit_sample.py</code> , SWHID checked with <code>/known/</code>	<code>samples/<pl>/<sha1_git>/</code>
Review a program	review UI, blinded; reviewer = human · tool · LLM	one immutable JSON per review
AI agents	parallel campaigns in git worktrees	one commit per language

Today: forms → GitHub issues → pull requests, accepted by default, reverted if wrong. **The UI and the workflow are clunky, and will be redesigned.**

What to keep

- ▶ **every contribution is a traced fact:** who, when, why, with which evidence
- ▶ **open to discussion:** each claim has a thread; disagreement is kept, not overwritten
- ▶ **revertible;** humans, tools and LLMs on the same footing
- ▶ facts as files, aggregates derived

Challenges: what's missing

Programming language identification (PLI) at archive scale: the central challenge

- ▶ **Coverage:** PLI tools know ~1,100 syntaxes (Linguist, Pygments, Synid), Tree-sitter several hundred grammars; we catalogue ~12,600 languages, 90% without an extension
- ▶ **Ambiguity:** 44 extensions have 6+ claimants (`.m`, `.h`, `.pl`, `.t`); ~70 with several primary claimants have no disambiguation rule
- ▶ **No ground truth:** no benchmark measures PLI tools on SWH, per language; reviews and extension studies can build one
- ▶ **Scale:** billions of files: cheap filters on the compressed graph, costly ones on content, every level validated (P / R / F1)

Identity

a canonical language ID authority, an "ORCID for languages"; merge `Python` / `py` / ...; bring HOPL's genealogy in

SWH-native evidence

per-extension **extractors** (every file + origin, from the SWH graph) are costly: CodeCommons tools, so far run for `.cbl`, `.fsf`, `.rpgle`; then a **complete mapping** of SWH programs (a laptop scan: killed after 6.8 h)

Evidence cards

for all ~12,600 languages, with a formal evidence-of-absence verdict

Governance & hosting

who curates? what counts as a language? data license? Today: one person's repo + GitHub Pages

How you can help, concretely

SWH / CodeCommons infrastructure

- ▶ run the CodeCommons extractors on more extensions (as for `.cbl`, `.fsf`, `.rpgle`), towards a complete mapping
- ▶ a per-extension index in the derived datasets
- ▶ graph / provenance access; a neutral home

PL / ML / MSR researchers

- ▶ PL identification at archive scale: Tree-sitter grammars, classifiers, a PLI benchmark built from SWH
- ▶ sampling & labeling methodology
- ▶ benchmarks, fine-tuning data for low-resource languages

Domain experts: adopt an extension

- ▶ COBOL, neuro `.fsf`, `.m`, IBM RPG... or yours
- ▶ dive deep: define the taxonomy, review, tell the story

Everyone, one click

- ▶ add a language · label an extension
- ▶ review a program · flag a hallucination

Catch them all!

Gamifying the encyclopedia: programs and languages

● COBOL 2 on the site · ≈150K in the study	● Perl caught · 6 programs	● Pascal caught · 5 programs	● Mercury caught · 1 program
○ C seen · not yet caught	○ Fortran 29 variants · none caught	○ Icon 5 catalogs · 0 programs	○ Smalltalk 6 sources · 0 programs
○ Brainfuck 6 sources · 0 programs	○ ALGOL 60 0 programs	○ Porth found by an LLM	✗ BASIC-D invented by an LLM

192 of 12,608 languages caught (1.5%). A **pipeline gap** more than an archive gap: extracting every file of an extension from SWH is costly; CodeCommons extractors did it for `.cbl`, `.fsf`, `.rpgle`, not yet for the others.

- ▶ **Catch** a language: attach its first verified program from SWH; every catch is reviewed, invented languages get caught out
- ▶ **Power-ups**: each per-extension extractor (CodeCommons) catches a whole extension at once: `.cbl/.CBL` alone yields ≈150,600 COBOL files
- ▶ **Rarity**: the fewer programs in the archive, the rarer the catch
- ▶ **Quests**: catch C and Fortran (still missing!), label `.flat`, check the 465 unverified LLM names
- ▶ **Collections**: a family (ALGOL's descendants), a decade, an extension
- ▶ **Leaderboards**: for people, and for models (the LLM game already counts 18)

Proposal: an official SWH × CodeCommons initiative

Start small, scale incrementally, on a reproducible pipeline

A reproducible build, not a hand-maintained site

- ▶ **inputs of any kind**: catalogs, HOPL, SWH datasets, PLI tools, LLM proposals and judgements, **human insights and reviews**
- ▶ one versioned pipeline turns them into the encyclopedia: site, datasets, evidence cards
- ▶ rebuilt from scratch at any time; every number traceable to its inputs

Step

What (to discuss)

1 · start small

neutral home · canonical language IDs · the reproducible build · 3–5 adopted extensions

2 · grow

SWH-native evidence (graph, derived datasets) · a PLI benchmark · evidence cards for all languages

3 · scale

archive-wide with CodeCommons: usage over time, papers · MSR paper (method) · data paper

Decisions for today

Who's in? · Where does it live? · Which extensions do we adopt first?

Towards an encyclopedia of programming languages anchored in Software Heritage & CodeCommons

Every language with its programs, every program with its language.

From the beginning of computing to today, with Software Heritage as the source of truth for the code.

PL Catalog: blog.mathieuacher.com/PL-ultimate-llm · github.com/acherm/PL-ultimate-llm

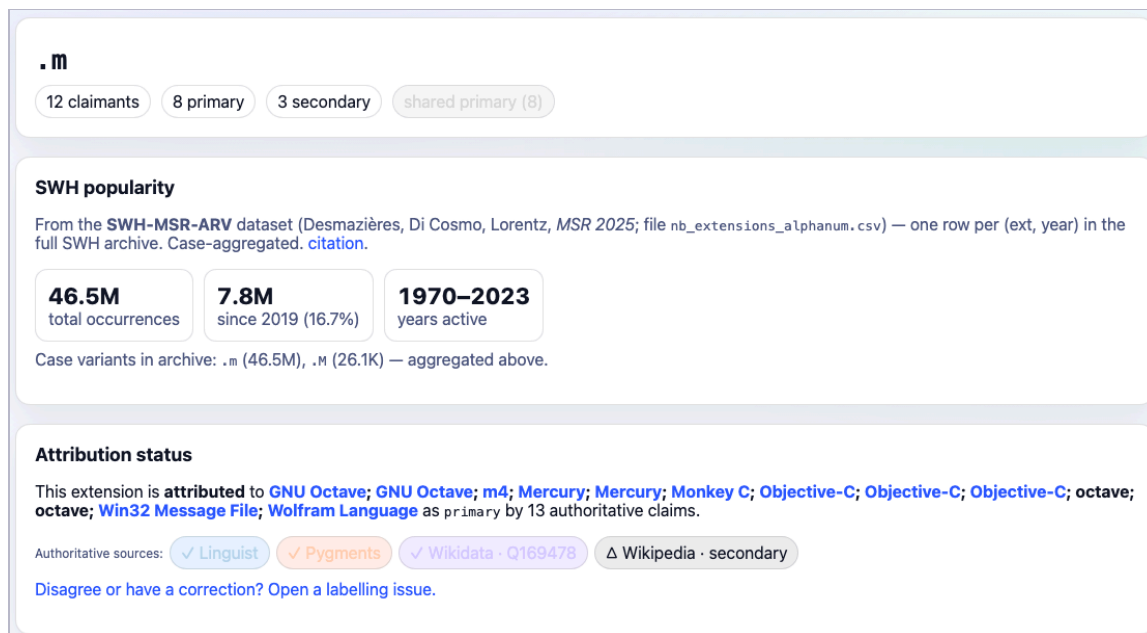
Reports: [docs/cobol_swh_study.pdf](#) · [fsf_swh_study.pdf](#) · [rpgle_swh_study.pdf](#)

Thanks to Software Heritage (Thomas Aynaud, Roberto Di Cosmo, Stefano Zacchiroli, Valentin Lorentz) for feedback, assistance and the SWH extractions; to the DiverSE team (Rennes) and the PLI group (Caroline Landry, Axel Amour N'cho, Baptiste Mehat, Bignon Lokonon, Corentin Ollivier, Guillaume Claudic, Stephan Kunne); and to everyone who will adopt an extension.



Backup

Backup · polysemy: an extension is not a language



- ▶ `.m`: **46.5 M** files, **12 claimants**
- ▶ Linguist's **377** disambiguation rules, made **runnable**: "Mercury, because `:- module` matched"
- ▶ Then `XSetModifierMapping.m` (Xorg) is... **none of them**: an xts5 test-case format
- ▶ `.icn` → Unicon, `.icon` → Icon, but Icon sources are `.icn`

Classify content. Keep the provenance of every claim.

For each language, Software Heritage is the **source of truth for the code**:

✓ Evidence of existence

- ▶ $\geq K$ **real archived programs**
- ▶ each with a **qualified SWHID**: citable, permanent
- ▶ and **why** it was attributed: extension? heuristic rule? classifier? human?

∅ Evidence of absence

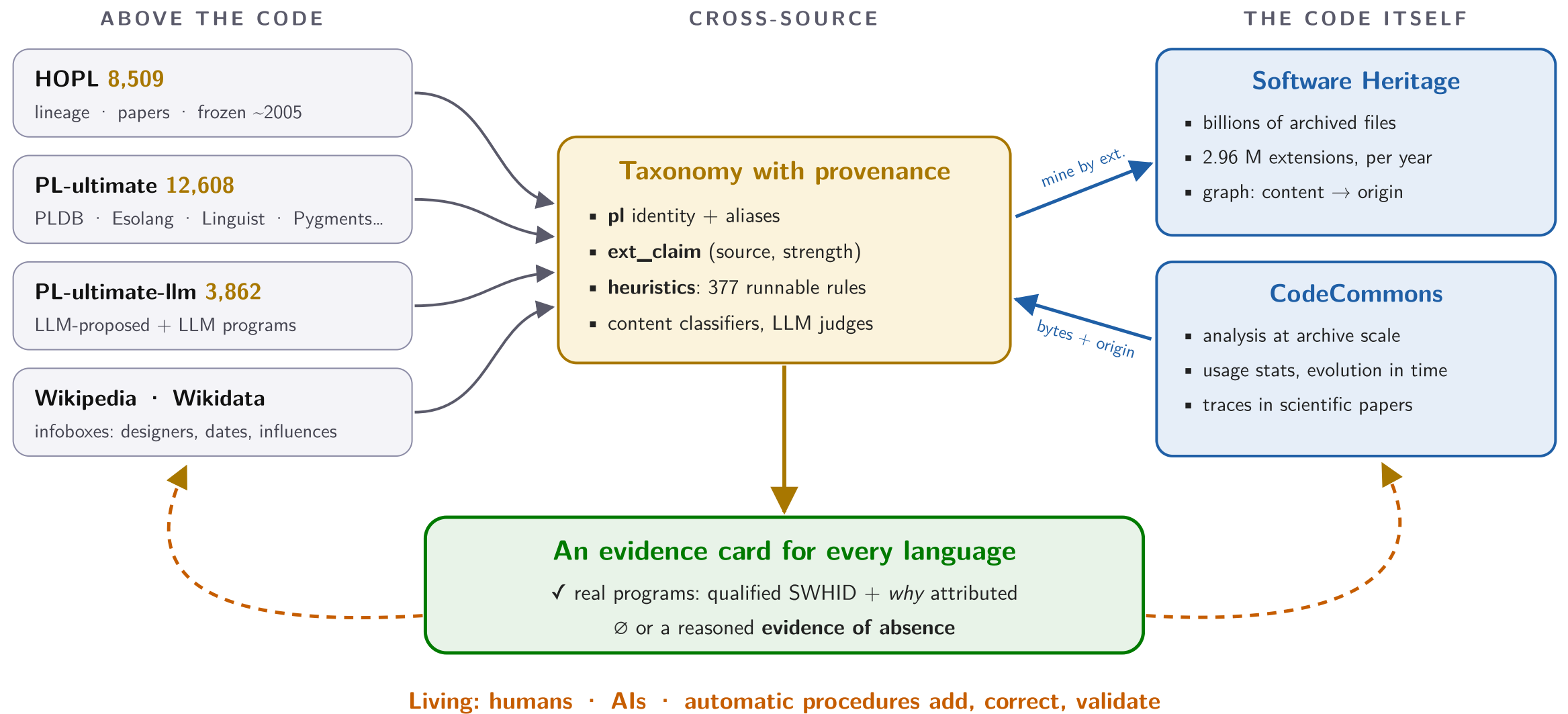
- ▶ no extension claimed anywhere
 - ▶ extension claimed, absent from SWH
 - ▶ files exist, but always won by a competitor (Mercury vs MATLAB on `.m`)
- A result in itself.

↻ Then, at CodeCommons scale

- ▶ usage statistics
- ▶ evolution over time: birth, peak, decline
- ▶ traces in scientific papers

Living: humans, AIs and automatic procedures add, correct, validate. Goal: **every language, from the beginning of computing to today**.

Backup · how the pieces fit



Backup · the landscape, source by source

Source	Size	Captures	Limitation
HOPL	~8,500	lineage, papers	frozen ~2005, often down, no code
Esolang	~6,800	esoteric, experimental	many with ~zero usage
PLDB	~5,100	paradigms, dates, designers	sparse on real adoption
Linguist	~800	detection + ambiguity rules	GitHub's worldview
Pygments	~600	syntax highlighting	exists if someone wrote a lexer
Rosetta Code	~560	tasks × languages	toy snippets
PL-ultimate	12,608	union of 7 (+ Wikidata)	88% single-source
PL-ultimate-llm	~3,900	LLM-proposed + programs	hallucination risk
SWH-MSR-ARV	2.96 M exts	occurrences per year	extensions, not languages

```
swh:1:cnt:c5ecce87432c6b1e8465269cb9b15902fe0dc302
;origin=https://github.com/st3fan/osx-10.9
;anchor=swh:1:rev:9f8135f87c2ba27b1e76a514464b823812effd45
;path=/CPANInternal-140/DBIx-Class/t/resultset_overload.t
```

- ▶ `.t` is shared by Perl, Raku, Terra, Turing → Linguist rule `h/linguist/.t/0` matched → **Perl**
- ▶ The bytes were fetched at the pinned commit and their `sha1_git` verified; a git commit SHA **is** an SWH revision ID
- ▶ Verification of all samples via SWH `/known/`: content **261/261**, anchor commit **251/261**, origin **233/261**
- ▶ **Gap**: the "seen $N \times$ in SWH" count comes from a filename + length match, not a strict content-id match; fixing it needs the nodes table (~840 GB) or an SWH-side index

Backup · the LLM contest protocol

```
turn: add Icon (+1 program)
```

```
List-Digest: 95f48d04
```

```
Model: meta-llama/llama-3.1-70b-instruct
```

```
Temperature: 0.4
```

```
turn: add Forge (+1 program)
```

```
List-Digest: d2538df5
```

```
Model: claude-sonnet-4-6
```

```
Agent: claude-code
```

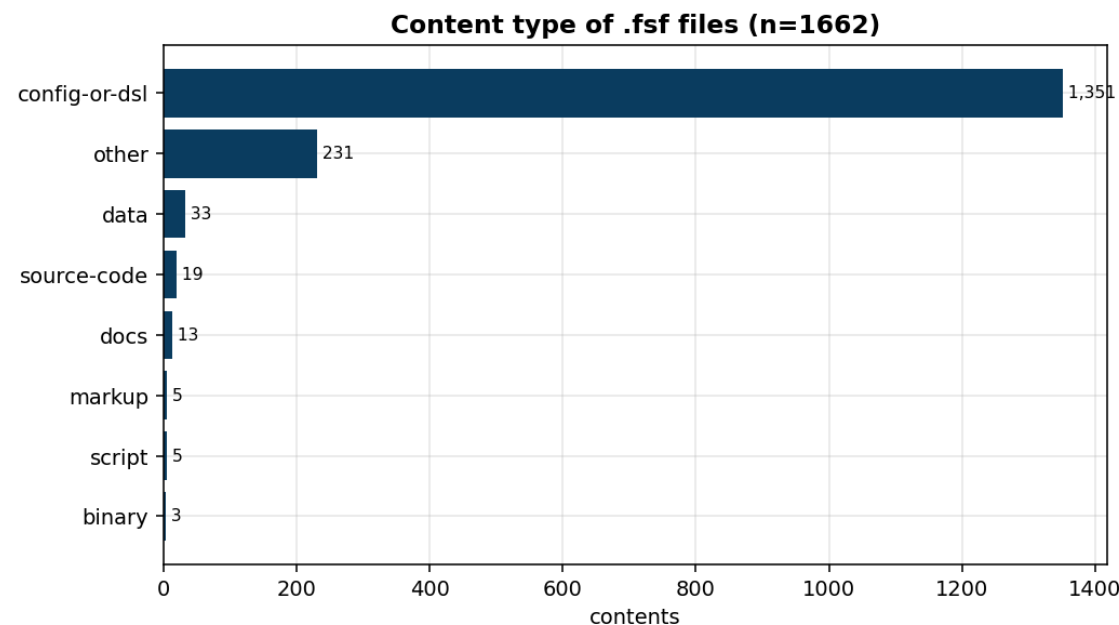
```
WebSearch: disabled
```

```
Strategy: batch-recall-100
```

- ▶ **List-Digest** pins the list the model saw; hooks reject duplicates (case-insensitive) and bad hashes
- ▶ 3,846 **turn: add** commits since Oct. 2025: Claude 3,392 · Grok 246 · Gemini 144 · Llama 48 · GPT/Codex 32 · DeepSeek 9 · Mistral 2
- ▶ Well-known languages are exhausted: agents now dig into regional languages, niche DSLs, historical systems

Backup · .fsf: an embedded DSL for neuroimaging

- ▶ **FSL FEAT** design files: fMRI analysis configuration
- ▶ a **declarative DSL** hosted in **Tcl** (`set fmri(...)` value)
- ▶ 1,662 files judged: **99%** configuration / DSL rather than general-purpose code; Tcl in 77%
- ▶ provenance: OpenNeuro, neuroimaging labs; top repo 18%
- ▶ polysemy tail: **GLSL fragment shaders**, git-annex / DataLad pointers
- ▶ a one-line marker (`set fmri()`) agrees with the judge **100%**



Backup · the review app

COBOL labels

browse

disagreements

unreviewed

with origin

dashboard

2783

labelled contents

2783

cached bytes

1

human-reviewed

2780

recovered origins

459

rule-labelled

is-COBOL label sources — Judge: the LLM verdict · reclassifier: deterministic content rules (reclassify.py) · oracle: LLM is-COBOL used to validate the reclassifier · division-gate: cheap baseline = has ≥2 COBOL divisions · human: your reviews. A file is Δ when its available is-COBOL votes disagree.

GROUP RULES

not-cobol:synthetic | origin = https://gitlab.com/tbetestpublic/repo-with-many-files-in-one-tree.git (459 contents) · WBC_*.FOO.CBL auto-generated placeholder (This is cobol file number N)

DATASETS

1k: 1000 | lo: 350 | main: 350 | pilot: 100 | eval:tail: 92 | eval:control: 50 | eval:wbc: 20

IS-COBOL AGREEMENT

judge vs reclassifier: 2080 agree · 1 disagree

reclassifier vs LLM-oracle: 143 agree · 19 disagree

RECLASSIFIER LABEL DISTRIBUTION

cobol	2080	
synthetic-placeholder	459	
other	122	
comic-book-list	45	
cobol-copybook	28	
binary-data	25	
cobol-generated	24	

One pane over every label for a file: LLM judge · reclassifier · gate · human · origin. Disagreements are surfaced; group rules label a whole origin at once.

.rpgle labels

browse

non-RPGLE

unreviewed

rule-labelled

/SAX2.rpgle

/SAX2.rpgle

swh:1:cnt:f687e99428d1d0bb8329ea4b129289f6921f05a · 13603 bytes · SWH_

* Summary: SAX2 parser interface used to build the DOM tree

* Description: those are the default SAX2 interfaces used by the library when building DOM tree.

*

* Copy: See Copyright for the status of this software.

*

* Author: Patrick Monnerat <pm@datasphere.ch>, DATASPHERE S.A.

/if not defined(XML_SAX2_H_)

/define XML_SAX2_H_

/include "libxmlrpg/xmlversion"

/include "libxmlrpg/parser"

/include "libxmlrpg/xlink"

d xmlSAX2GetPublicId...

d pr * extproc('xmlSAX2getPublicId') const xmlChar *

d ctx * value void *

d xmlSAX2GetSystemId...

d pr * extproc('xmlSAX2getSystemId') const xmlChar *

d ctx * value void *

d xmlSAX2SetDocumentLocator...

d pr * extproc('xmlSAX2SetDocumentLocator')

d ctx * value void *

d loc value like(xmlSAXLocatorPtr)

d xmlSAX2GetLineNumber...

d pr 10i 0 extproc('xmlSAX2GetLineNumber')

d ctx * value void *

d xmlSAX2GetColumnNumber...

d pr 10i 0 extproc('xmlSAX2GetColumnNumber')

d ctx * value void *

d xmlSAX2IsStandalone...

d pr 10i 0 extproc('xmlSAX2IsStandalone')

d ctx * value void *

d xmlSAX2HasInternalSubset...

d pr 10i 0 extproc('xmlSAX2HasInternalSubset')

d ctx * value void *

ORIGIN

<https://github.com/kiyolee/libxml2> | github.com

LLM JUDGE

content_type	copybook-or-header
language	RPG IV / ILE RPG (fixed-format)
source_format	fixed-format
unit_kind	copybook-prototype-header
related_languages	RPG, C
embedded SQL	False
platform	ibm-i-as400
domain / maturity	library-framework · library-quality
is PL?	True

Prototype header (copybook) for the libxml2 SAX2 parser interface, providing RPG IV procedure prototypes (PR definitions with EXTPROC) that bind to the C-language xmlSAX2* functions in libxml2. Intended to be /include-d by RPG programs that need to call the SAX2 DOM-building callbacks. Uses conditional compilation directives to conditionally expose SAX1-compatible and HTML/DOCB handler prototypes.

INDICATORS

lines	248
comment ratio	0.034
**FREE	False
format guess	fixed-format
fixed spec lines	175
/free blocks	0
dol- / dol-proc / dol-pr	0 / 0 / 0
EXEC SQL	False
/copy /include	3
looks copybook	True
reclass	rpgle-copybook

ASSERT FOR A GROUP

One statement labels every matching content (e.g. all files from a non-RPGLE repo).

Scope

this origin

Value

<https://github.com/kiyolee/>

Label

not-rpgle

Buttons

```
{
  "schema": 1,
  "subject": { "sha1_git": "f851a314...", "filename": "XSetModifierMapping.m", "ext": ".m" },
  "reviewer": { "kind": "human", "id": "mathieu-acher" },
  "verdict": { "label": "pl/new:tet-scenario", "confidence": "high", "supersedes": null },
  "comment": "xts5 TET 'mc' format: C embedded in >>CODE blocks.",
  "shown": { "claimants": ["pl/m4", "pl/matlab", "pl/mercury", "..."] },
  "created_at": "2026-06-12T09:31:02Z"
}
```

- ▶ `reviews/<sha1_git>/<stamp>--<reviewer>--<hash>.json`: one immutable file per review, never a merge conflict
- ▶ reviewers are **humans, tools or LLMs**; what the UI showed is recorded (anchoring-bias trace)
- ▶ consensus is **derived**, and can be recomputed under other rules